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1507 1992

"On my honor, I have neither given nor received any unauthorized and/or inappropriate asistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues."

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1) $\begin{bmatrix} 1 & \alpha & -y & \alpha \\ 3\alpha & 5 & -9y & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1/2 & 1 \end{bmatrix}$

a) $\alpha = 0$
$y = 0$
$\beta = 1$
$k = 0$
$m = 0$

b) $R_3 = R_3 + R_2$
for upper triangle

c) $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

2) A geometry in "k" dimensional space is given by $x = 5y$ expression, where "x" and "y" represents the axis. The intersection with another geometry give a volume.

a) Give the minimum dimension of the space with your valuable comments.

b) Propose an expression for the unknown geometry to have a volume in intersection.

c) I proposed a geometry $x + 4y - z = 6$. Find all vectors in the intersection. Do these vectors form a vector space? The intersection has how many dimensions inside how many dimensional space?

3) Suppose u is a unit vector in R^n. This problem is about the n by n symmetric matrix $H = I - 2uu^T$

a) Show directly that $H^2 = I$

b) Show that H is orthonormal matrix.

c) One eigenvector of H is u itself. Find the corresponding eigenvalue.

d) H is diagonalizable.

4) Two tanks are connected, first tank has $v(0)$ 10 liters liquid and the second tank $w(0) = 40$ liters of water. The liquid diffusion between the tanks are proportional to the difference $v - w$.

a) Which one of the following differential equation set expresses the process?

$$\begin{vmatrix} V_{k+1} \\ \omega_{k+1} \end{vmatrix} = \begin{vmatrix} -2 & 1 \\ 1 & -2 \end{vmatrix} \begin{vmatrix} V_k \\ \omega_k \end{vmatrix} \qquad \frac{d}{dt}\begin{bmatrix} v \\ \omega \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} v \\ \omega \end{bmatrix} \qquad \frac{d}{dt}\begin{vmatrix} u \\ w \end{vmatrix} = \begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} u \\ w \end{bmatrix}$$

b) Determine percentages of the mixture in tank1 at time $t = 1$ sec.

c) Find how much liters of liquids are exist in each tank in steady state?